Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	27 June 2026
Project Name	OptiCrop
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Input	Enter Nitrogen (N), Phosphorus (P), and Potassium (K) values. Enter pH, temperature, humidity, and rainfall values.
FR-2	Data Validation	Validate all input values. Display an error message for invalid or missing inputs.
FR-3	Crop Recommendation	Process the input data. Predict the most suitable crop using the machine learning model. Display the recommended crop.
FR-4	Result Display	Show the crop recommendation to the user. Provide the prediction clearly on the user interface.
FR-5	Dataset Management	Load the crop dataset. Preprocess the dataset for prediction.

FR-6	Machine Learning Model	Train the prediction model. Use the trained model for crop recommendation.
FR-7	System Management	Store the trained model. Allow future updates with new datasets and improved prediction models.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	**Non-Functional Requirement**	**Description**
NFR-1	Usability	The application should provide a simple and user-friendly interface that allows farmers to enter data and receive crop recommendations easily.
NFR-2	Security	The system should validate user inputs and securely process data to prevent invalid or unauthorized access.
NFR-3	Reliability	The application should generate consistent and accurate crop recommendations based on the provided input data.
NFR-4	Performance	The system should process user inputs and display crop recommendations within a few seconds.
NFR-5	Availability	The application should be available whenever the local system or server is running and accessible to users.
NFR-6	Scalability	The system should support future enhancements, such

		as adding more crop datasets, prediction models, and cloud deployment.